

Supporting Information for: Microbial driver of 2006 – 2023 CH₄ growth indicated by trends in atmospheric δ D-CH₄ and δ^{13} C-CH₄

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Supporting Information Text

1. Shale Gas Impact on Fossil Fuel Source Signatures

The United States dominates global shale gas production, though China, Argentina, and Canada have ramped up production in recent years (*eia.gov*). Following (1), we compiled conventional ONG and shale gas CH₄ isotopic measurements for each country and flux-weighted them using production rate data (*eia.gov*) to determine the mean ONG source signature and its temporal variability. We found that China's ONG $\delta\text{D-CH}_4$ and $\delta^{13}\text{C-CH}_4$ became more enriched by 0.85‰ and 5.7‰, respectively, from 2007 to 2022 due to increased shale gas production data (Fig. S4c). Canada's ONG $\delta^{13}\text{C-CH}_4$ increased by 2.5‰ across the same time period. No estimate for changes in ONG $\delta\text{D-CH}_4$ due to shale gas could be made for Argentina due to a lack of isotopic data. For the United States, there is sufficient isotopic data to flux-weight US ONG $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ by basin-level conventional and shale gas ONG production data (1). Despite incorporating additional data, our results align with those of (1), with US ONG $\delta^{13}\text{C-CH}_4$ increasing by 2‰ between 2007 and 2022 (Fig. S4b). Our expanded analysis of US ONG $\delta\text{D-CH}_4$ reveals $\pm 5\%$ fluctuations but with little net change over the same period (Fig. S4a). For countries with no shale gas production, we assume constant ONG and coal source signatures over time.

2. Fossil Fuel Source Signature Flux-weighting Method Sensitivity

Our uncertainties in global mean fossil fuel source signatures do not consider uncertainties in the spatiotemporal distribution of emissions used in the flux-weighting (methods section 2.1). To test this effect, we flux-weighted our fossil fuel source signature maps (Fig. S3a-d) using posterior CT-CH₄ 1×1° gridded emissions and compared the global mean with the EDGAR8 emission inventory flux-weighting scheme (methods section 2.1). Because CT-CH₄ does not partition between coal and ONG emissions in the posterior (2), the prior (EDGAR8) grid cell-level partitioning was used to calculate the fossil source signature of each grid cell before global flux-weighting. We found that flux weighting with CT-CH₄ emissions reduces fossil fuel $\delta\text{D-CH}_4$ and $\delta^{13}\text{C-CH}_4$ by approximately 5‰ and 1‰, respectively, while preserving trends in the global mean after 2002 (Fig. S5). This shift is mainly driven by a reduction in East Asian emissions in the CT-CH₄

posterior, where fossil source signatures are enriched in ^{13}C and D (2). We incorporated these alternate global mean source signatures into our isotope mass balance calculations (equations 9 – 11) and found that calculated total fossil emissions are $<10 \text{ Tg yr}^{-1}$ higher and remain within the uncertainty bounds, and post-2006 emission trends remain unchanged (Fig. S11). This underscores the advantage of $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ records in constraining emission trends more effectively than total emissions. Flux-weighting uncertainties are considered in global mean microbial and biomass burning source signature calculations (Fig. S8; methods section 2.2 – 2.3).

Supporting Figures:

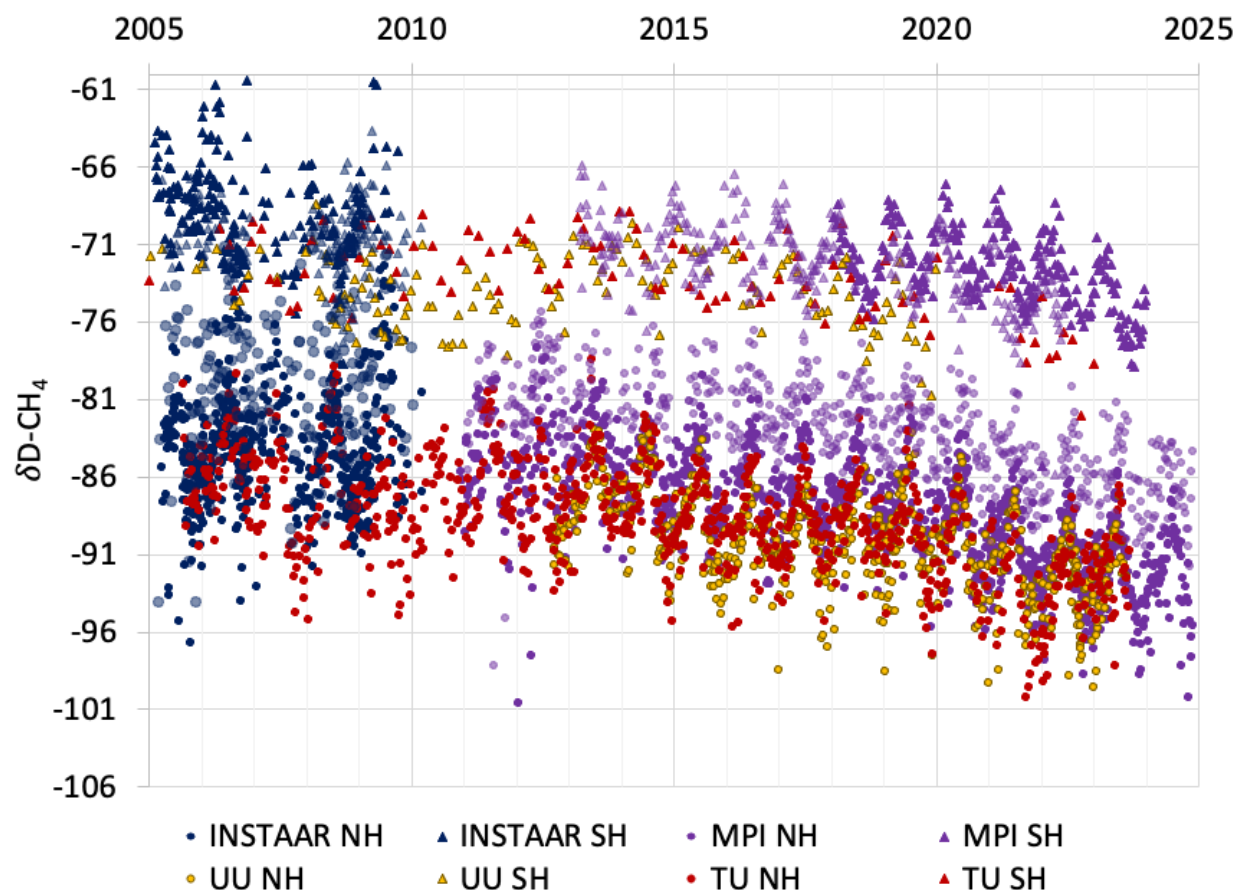


Fig. S1. Atmospheric $\delta D-CH_4$ Measurements. Marine Boundary Layer measurements of $\delta D-CH_4$ from INSTAAR (dark blue), MPI-BGC (purple), Utrecht University IMAU (yellow), and TU/NIPR (red) and the BUDS project (light blue). All data are harmonized to the MPI scale (Table S1) (3). NH and SH measurements are displayed as circles and triangles, respectively. Data from tropical sampling sites ($< 30^\circ N$ and $< 30^\circ S$) are shown as faded points.

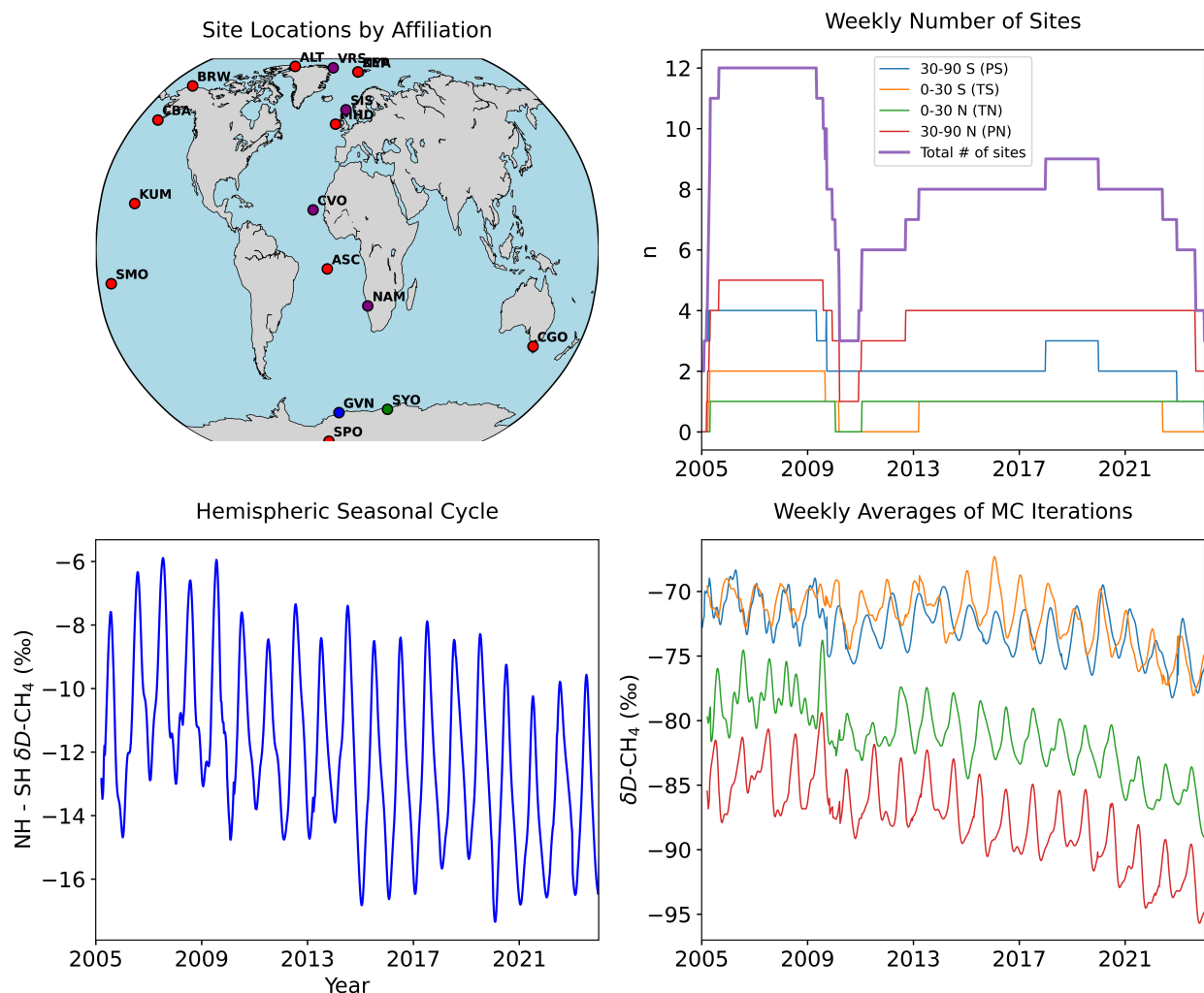


Fig. S2. Semi-hemispheric $\delta D-CH_4$ and seasonal cycle. **(Top Left)** World map showing the locations of MBL sites used in the global mean $\delta D-CH_4$ calculations. Sites are color coded based on the lab that produced the data (Purple = MPI, Blue = IMAU, Green = NIPR, Orange = INSTAAR). Note that several labs share site locations: IMAU and NIPR share NYA/ZEP, and INSTAAR and MPI share ALT, and MPI and IMAU share GVN. **(Top Right)** Number of measurements available in a given week from 2005 to 2023. Total number of measurements (purple), 30 – 90°N (red), 0 – 30°N (orange), 0 – 30°S (green) and 30 – 90°S (blue). **(Bottom Right)** Weekly semi-hemispheric average $\delta D-CH_4$ (same color scheme as A). **(Bottom Left)** Weekly interhemispheric difference (0 – 90°N minus 0 – 90°S).

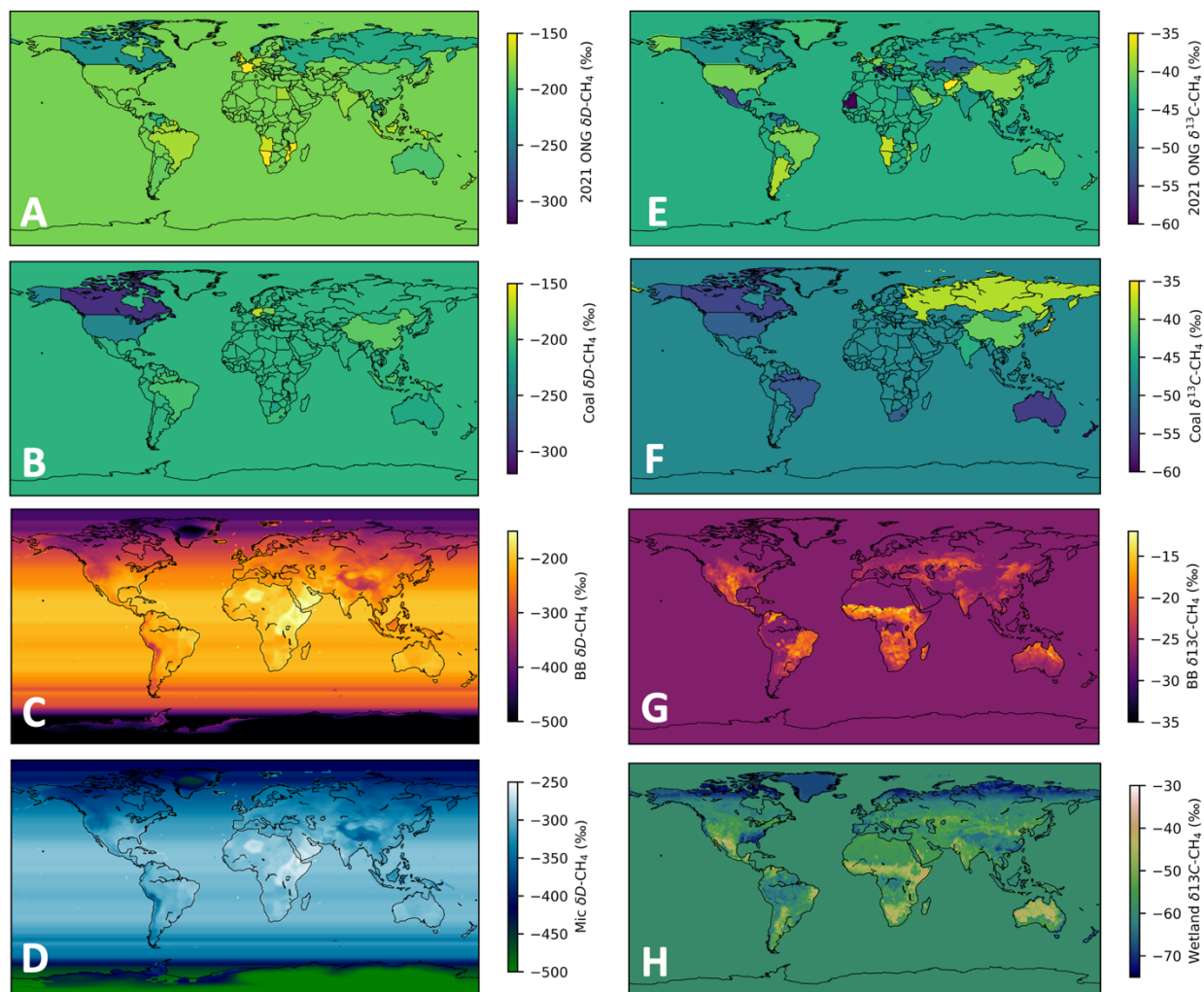


Fig. S3. 1°x1° gridded Source Signature Maps. $\delta\text{D-CH}_4$ (A - D) and $\delta^{13}\text{C-CH}_4$ (E - H) source signature maps. **(A & E)** Country-specific oil and natural gas (ONG) CH_4 emission source signatures. The United States, China, and Canada source signatures vary temporally due to increased shale gas production and associated emissions (Fig. S4). **(B & F)** Country-specific coal CH_4 emission source signatures. **(C & G)** Pyrogenic (Biomass burning, BB) CH_4 emission source signatures. **(D)** Microbial emission $\delta\text{D-CH}_4$. An alternative map was developed specifically for landfill emissions (not shown). **(H)** Wetland $\delta^{13}\text{C-CH}_4$ map (4). Note that although many grid cells have no CH_4 emissions (i.e., oceans), a theoretical value was provided to aid global 3D CH_4 modeling usage.

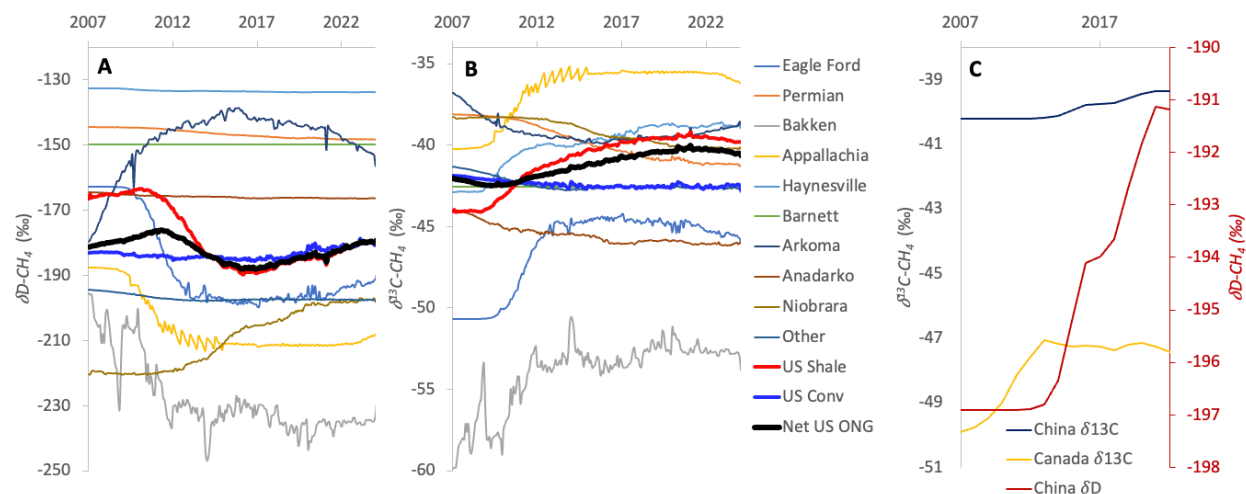


Fig. S4. Shale gas influence on US, Canada and China ONG Source signatures. Trends in United States geologic basin-specific annual mean ONG $\delta D-CH_4$ (A) and $\delta^{13}C-CH_4$ (B) following Lan et al., 2021. Flux-weighted mean US source signatures for Shale (red), Conventional ONG (blue), and all ONG (black) are shown. (C) Annual mean Canada ONG $\delta^{13}C-CH_4$ (yellow), China ONG $\delta^{13}C-CH_4$ (blue) and China ONG $\delta D-CH_4$ (red). No estimates were made for Canadian ONG $\delta D-CH_4$ due to lack of measurements. All trends are incorporated into global mean fossil fuel CH_4 emission source signature calculations.

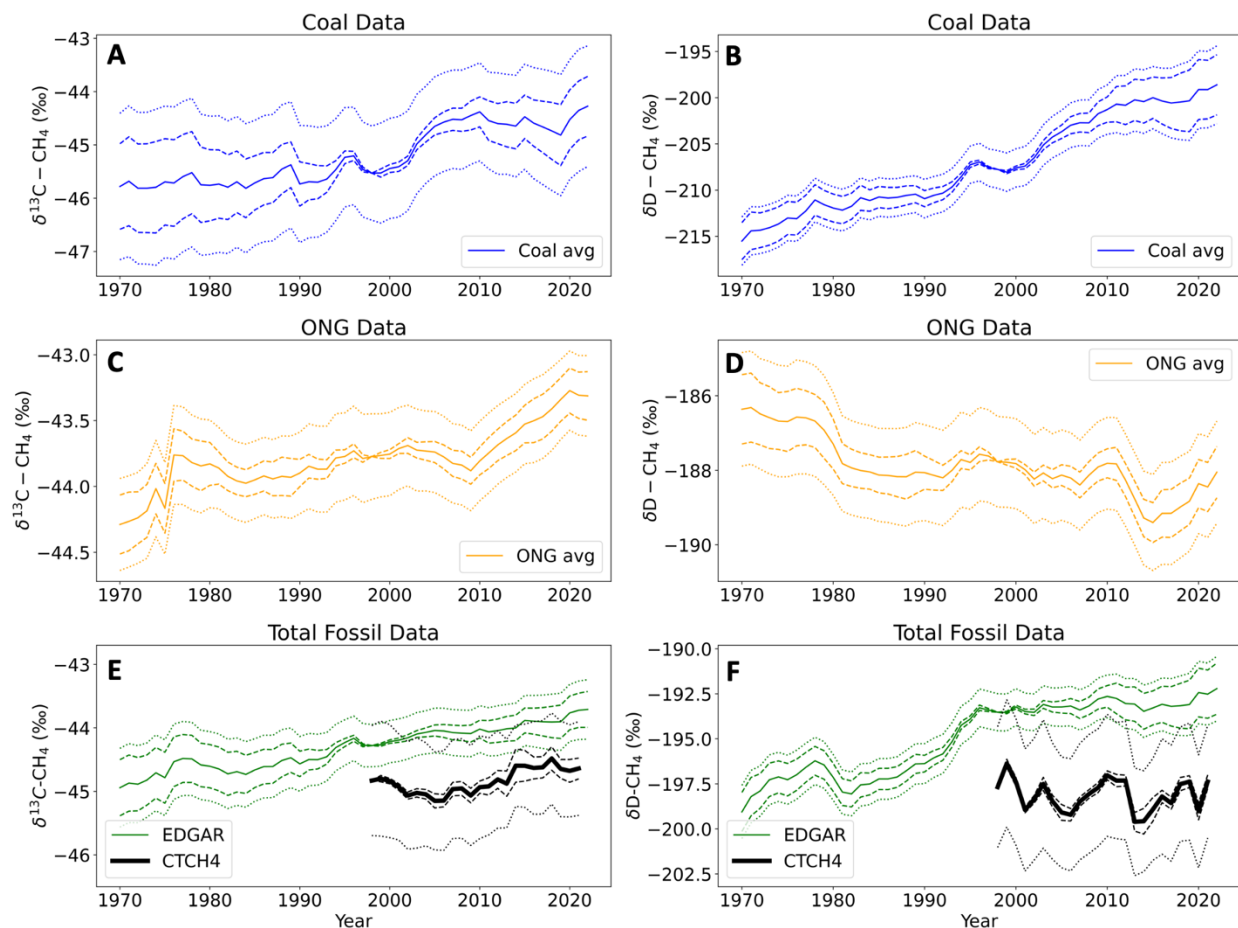


Fig. S5. Global annual mean fossil fuel source signatures and uncertainty. Global annual mean coal (Blue; A, B), ONG (Yellow; C, D) and total fossil fuel (Green; E, F) source signatures for $\delta^{13}\text{C}-\text{CH}_4$ (A, C, & E) and $\delta\text{D}-\text{CH}_4$ (B, D, & F) calculated by weighting Fig. S3 source signature maps with the EDGAR V8 emission inventory. Total two-sigma uncertainty (dotted lines) and two-sigma uncertainty relative to 1998 (dashed lines). Global annual mean fossil fuel $\delta^{13}\text{C}-\text{CH}_4$ and $\delta\text{D}-\text{CH}_4$ calculated by flux-weighting with CT- CH_4 posterior emission fields are shown in black (E, F).

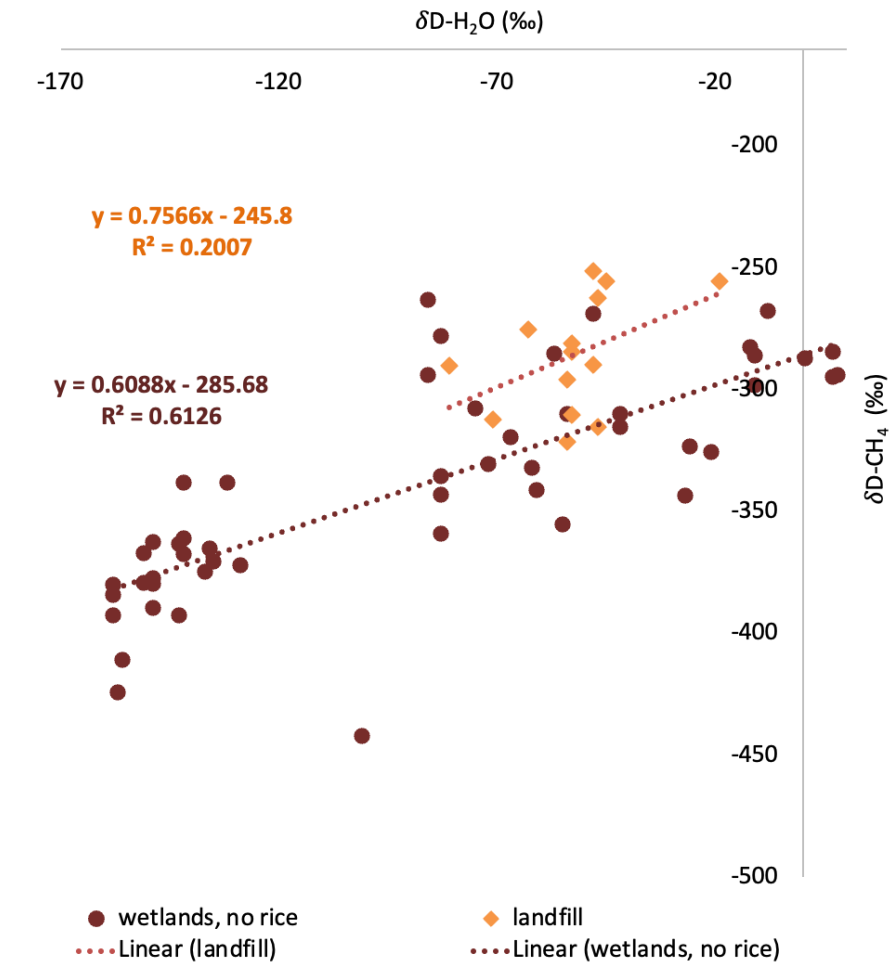


Fig. S6. Microbial $\delta\text{D-CH}_4$ regression comparisons. Comparison of the relationship between $\delta\text{D-CH}_4$ and $\delta\text{D-H}_2\text{O}$ for landfill CH_4 emissions (orange) and wetland CH_4 emissions (maroon). Wetland data is from (5), and the landfill data is from our updated compilation (Table S2) (*DOI for updated database will be published with manuscript*). The local $\delta\text{D-H}_2\text{O}$ was estimated using the global distribution of $\delta\text{D-H}_2\text{O}$ from (6). Though uncertain, the similar slope and enriched intercept of the landfill regression is consistent with expectations (5).

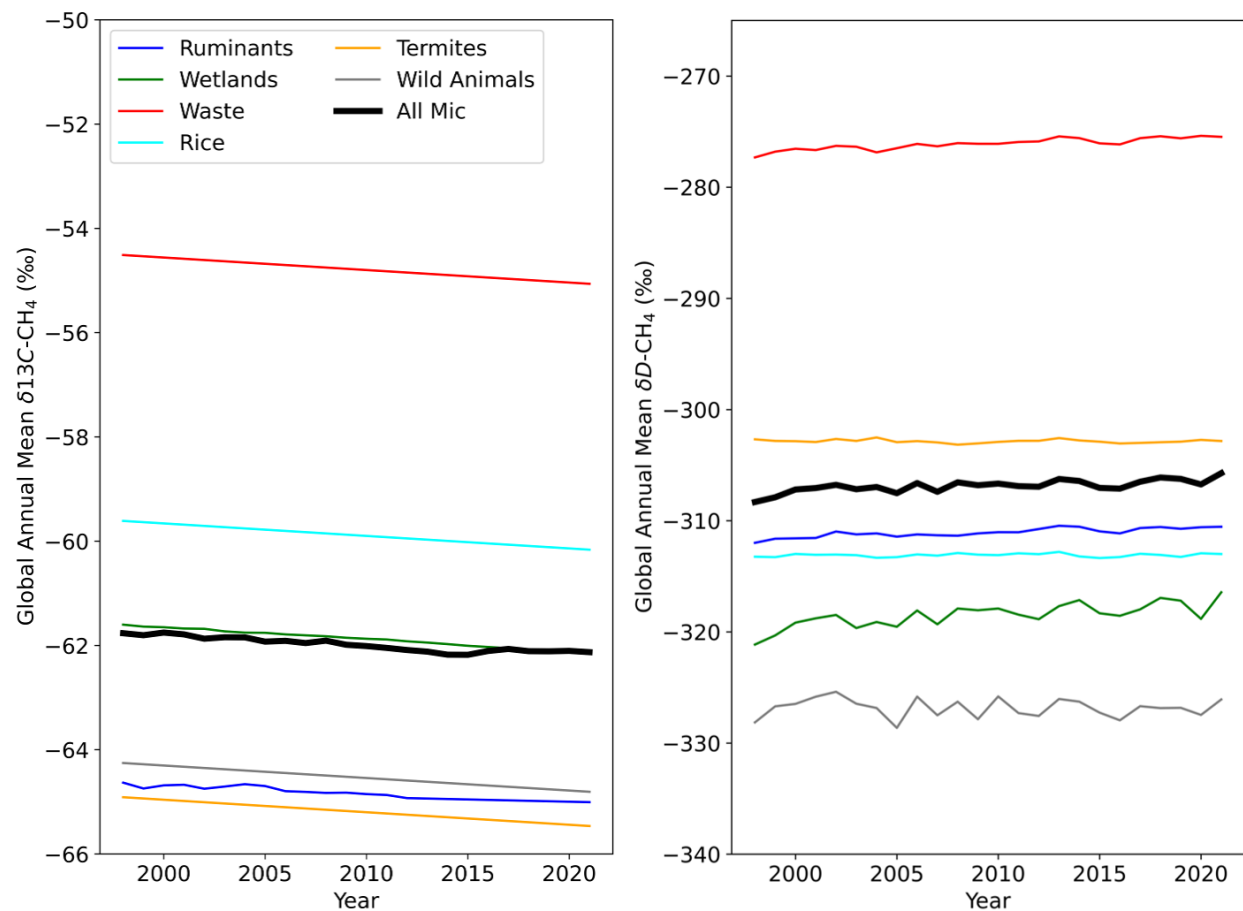


Fig. S7. Global annual mean microbial $\delta\text{D-CH}_4$ and $\delta^{13}\text{C-CH}_4$. Global annual flux-weighted mean microbial (black) and subcategory (color-coded) $\delta^{13}\text{C-CH}_4$ (left) and $\delta\text{D-CH}_4$ (right). The global mean ruminant source signature was extended to 2022 by extrapolating the linear decreasing trend identified in the study beginning in 2012 (7).

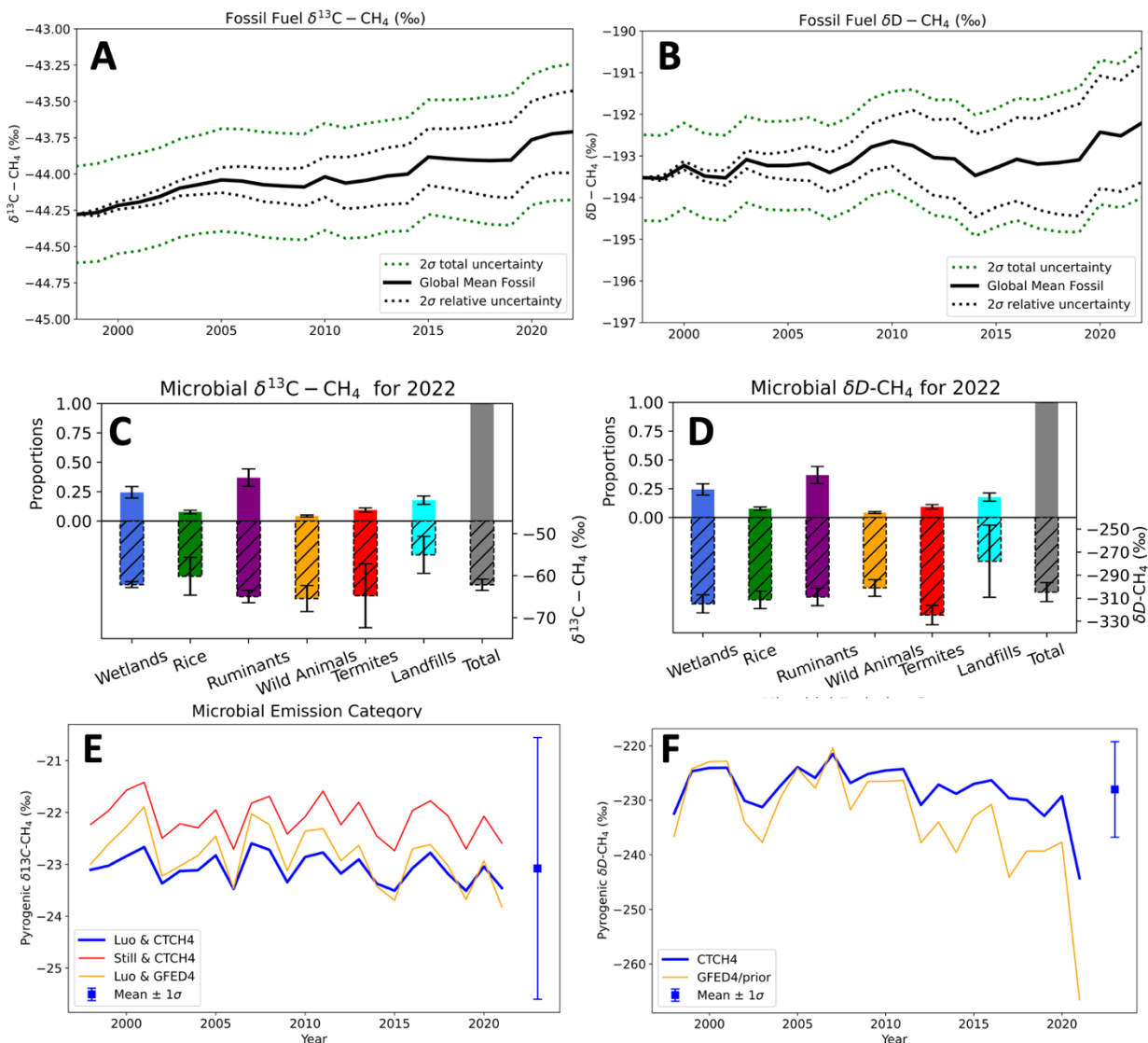


Fig. S8. Global annual mean source signatures and uncertainty. **(A)** Global mean $\delta^{13}\text{C} - \text{CH}_4$ of fossil fuel CH_4 emissions (solid black line). Total two-sigma uncertainty (green dotted line) and two-sigma uncertainty relative to 1998 (black dotted line). **(B)** Same as (A) but for $\delta\text{D} - \text{CH}_4$. **(C)** Global mean microbial emission subcategory $\delta^{13}\text{C} - \text{CH}_4$ (bottom dashed bar) and each subcategory's contribution to total microbial emissions (top solid bar) and 1σ uncertainties (see table S4 for values). Subcategories are color-coded and the net global average microbial $\delta^{13}\text{C} - \text{CH}_4$ and 1σ uncertainty is shown in gray. Results are shown for the year 2022. **(D)** Same as (C) but for $\delta\text{D} - \text{CH}_4$. **(E)** Global annual mean pyrogenic $\delta^{13}\text{C} - \text{CH}_4$ for each year calculated by flux-weighting the source signature map using GFEDv4.1 emissions (8) and C3/C4 data from (9) (yellow), CT- CH_4 posterior emissions and C3/C4 data from (10) (red) and CT- CH_4 posterior emissions and C3/C4 data from (9) (blue). Note the good agreement between the prior and posterior flux-weighting methods, but more negative values using the more recent C3/C4 map (blue line) (9). The CT- CH_4 mean and standard deviation of all years and MC simulations is shown on the right of the subplot. **(F)** same as (E) but for $\delta\text{D} - \text{CH}_4$, with results from flux-weighting with GFED4.1 (yellow) and CT- CH_4 (blue).

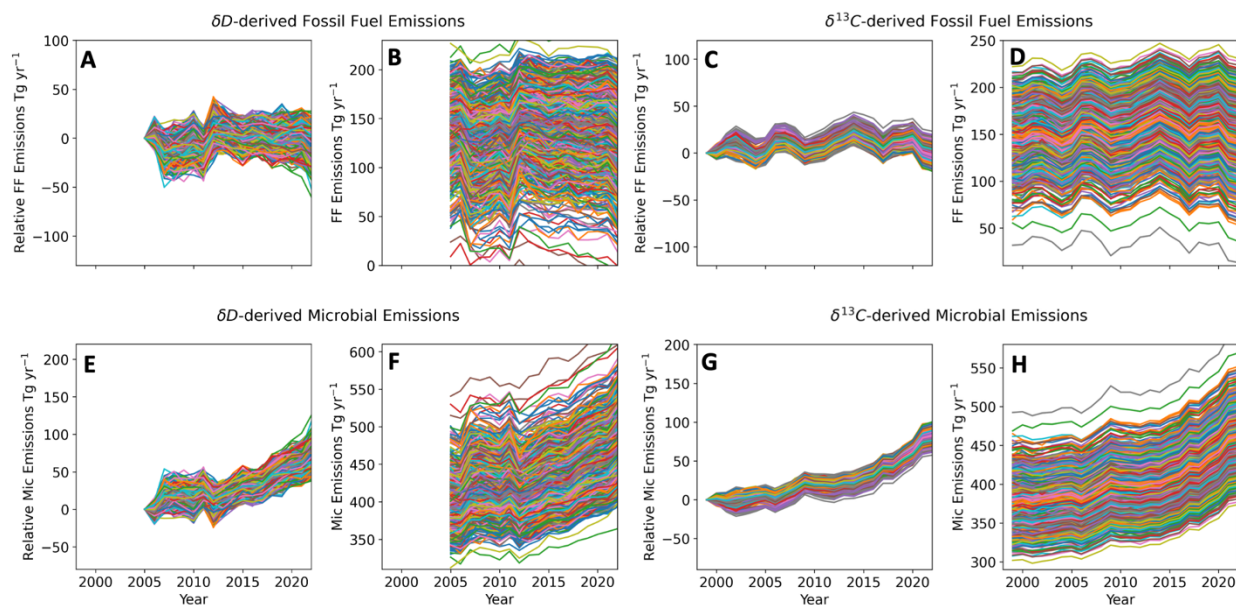


Fig. S9. Spaghetti diagrams of mass balance fluxes. Monte Carlo spaghetti diagrams show the uncertainty range of trends in global annual emissions by plotting the emissions relative to 1998 or 2005 (A, C, E, G) and both uncertainty in the trends and mean values of global annual emissions by plotting total emissions (B, D, F, H). (A, B) δD -derived Fossil Fuel emissions (C, D) $\delta^{13}C$ -derived Fossil Fuel emissions, (E, F) δD -derived Microbial emissions, (G, H) $\delta^{13}C$ -derived Microbial emissions.

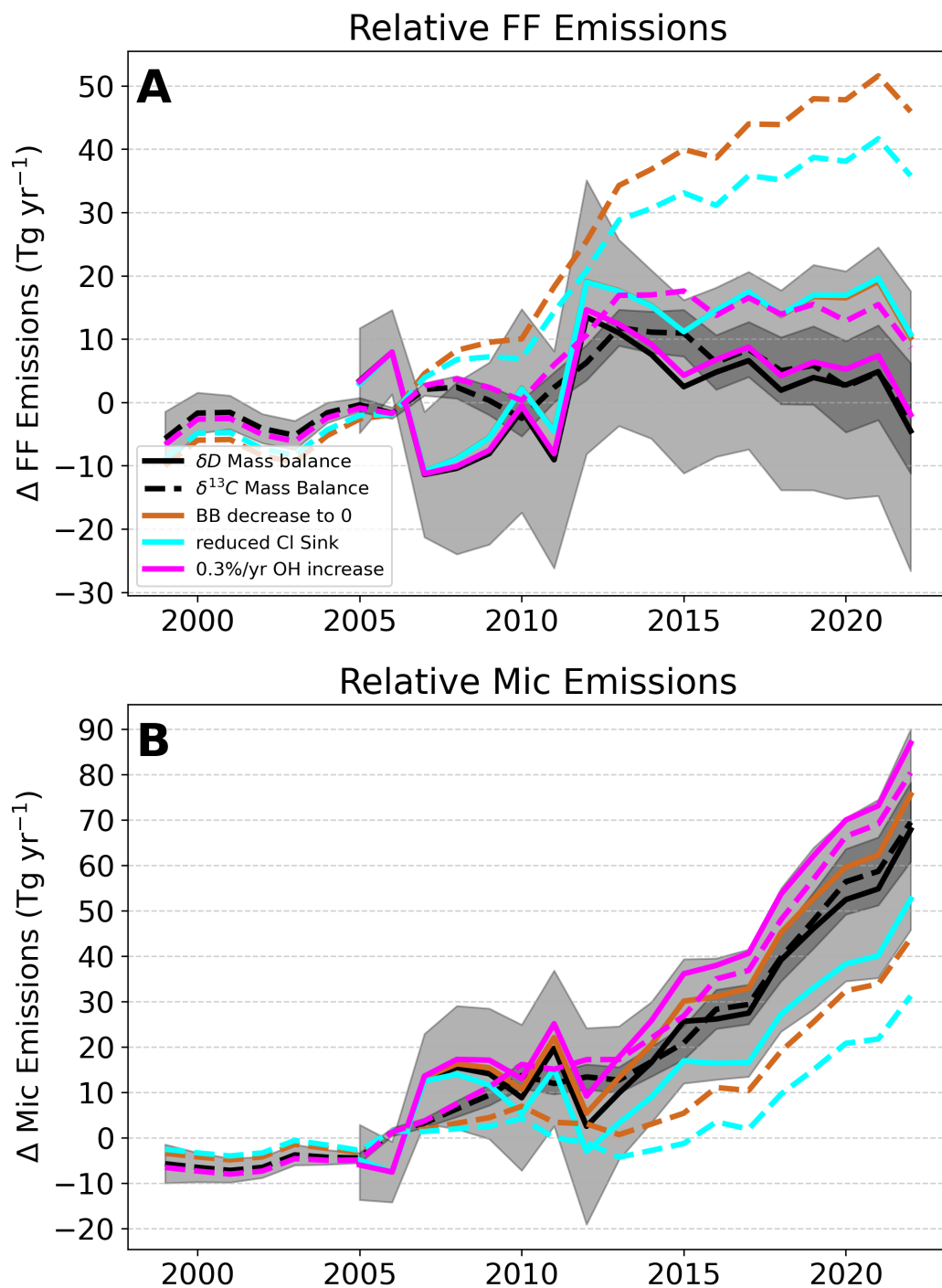


Fig. S10. FF and microbial emission estimates from changing BB Emissions, CI Sink, and OH sink. Relative Fossil Fuel (**A**) and microbial (**B**) emissions for the $\delta\text{D-CH}_4$ (solid lines) and $\delta^{13}\text{C-CH}_4$ (dashed lines) mass balances. Baseline scenario two-sigma error windows are shaded. The mass balance results for a 100% reduction in pyrogenic emissions from 2005 to 2022 are shown in brown. The mass balance results for a reduction in the CI sink from 3.5% to 1.0% of the total sink from 2005 to 2022 are shown in cyan. The mass balance results for an increase in the OH sink from recent estimates (11, 12) from 2005 to 2022 are shown in pink.

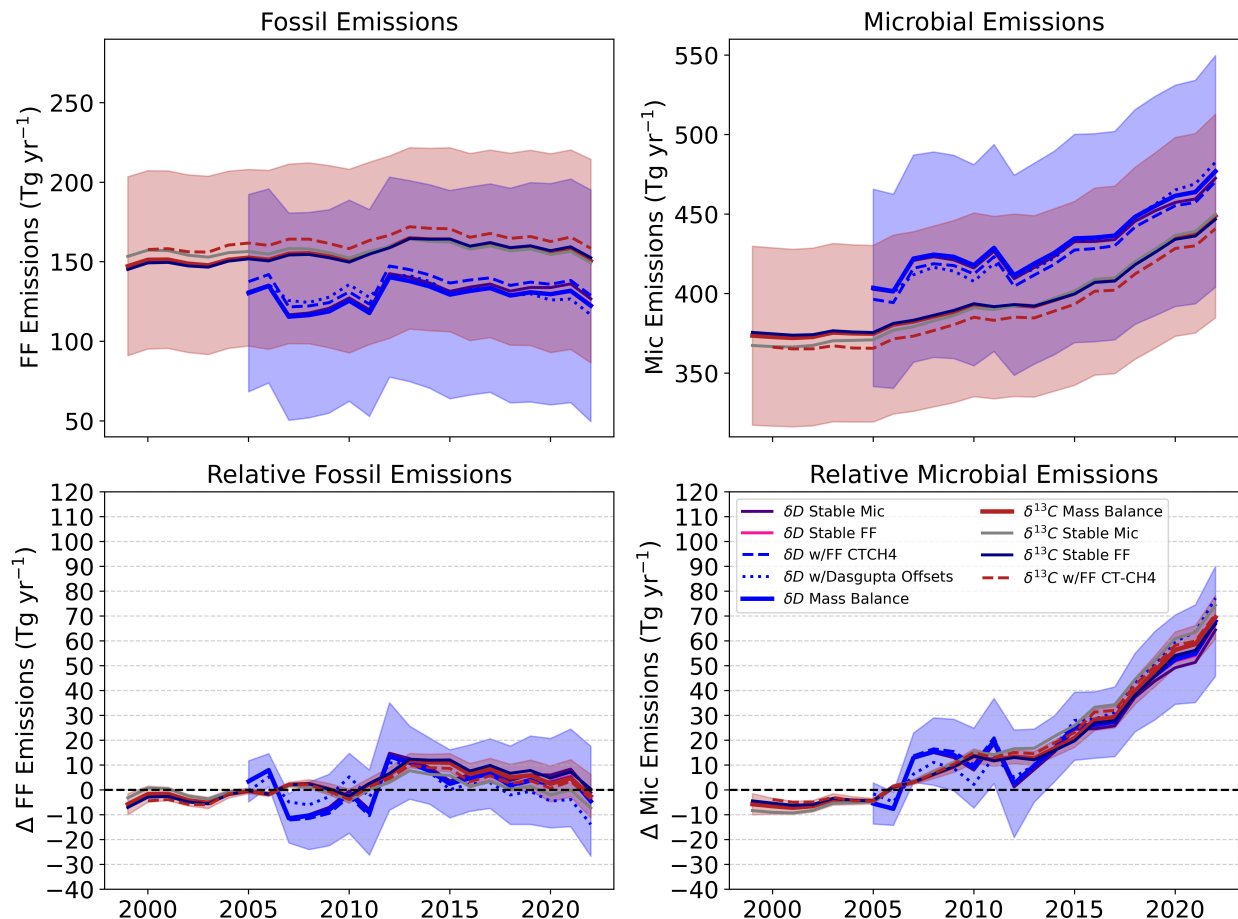


Fig. S11. Source signature sensitivity tests. **(A)** Global fossil fuel emissions. **(B)** Global microbial emissions. **(C)** Change in fossil fuel emissions relative to the 2005 – 2007 average. **(D)** Change in microbial emissions relative to the 2005 – 2007 average. Best-guess results are shown for δD -CH₄ (blue) and $\delta^{13}C$ -CH₄ (maroon) with two-sigma uncertainty shading. Additional scenarios with no uncertainty windows are as follows: Mass balance results calculated with stable microbial δD -CH₄ (purple), stable microbial $\delta^{13}C$ -CH₄ (gray), stable fossil fuel δD -CH₄ (pink), stable fossil fuel $\delta^{13}C$ -CH₄ (dark blue). Dashed maroon and blue lines display mass balance results calculated using global mean fossil fuel source signatures flux-weighted with CT-CH₄ posterior emissions as opposed to EDGAR inventories like in the baseline scenario. Dotted blue lines display mass balance results calculated using global mean δD -CH₄ that was determined using lab offsets from ref (13) (Table S1). Some lines are difficult to see because of overlap with the baseline modeling scenario, highlighting the minimal impact of certain sensitivity tests.

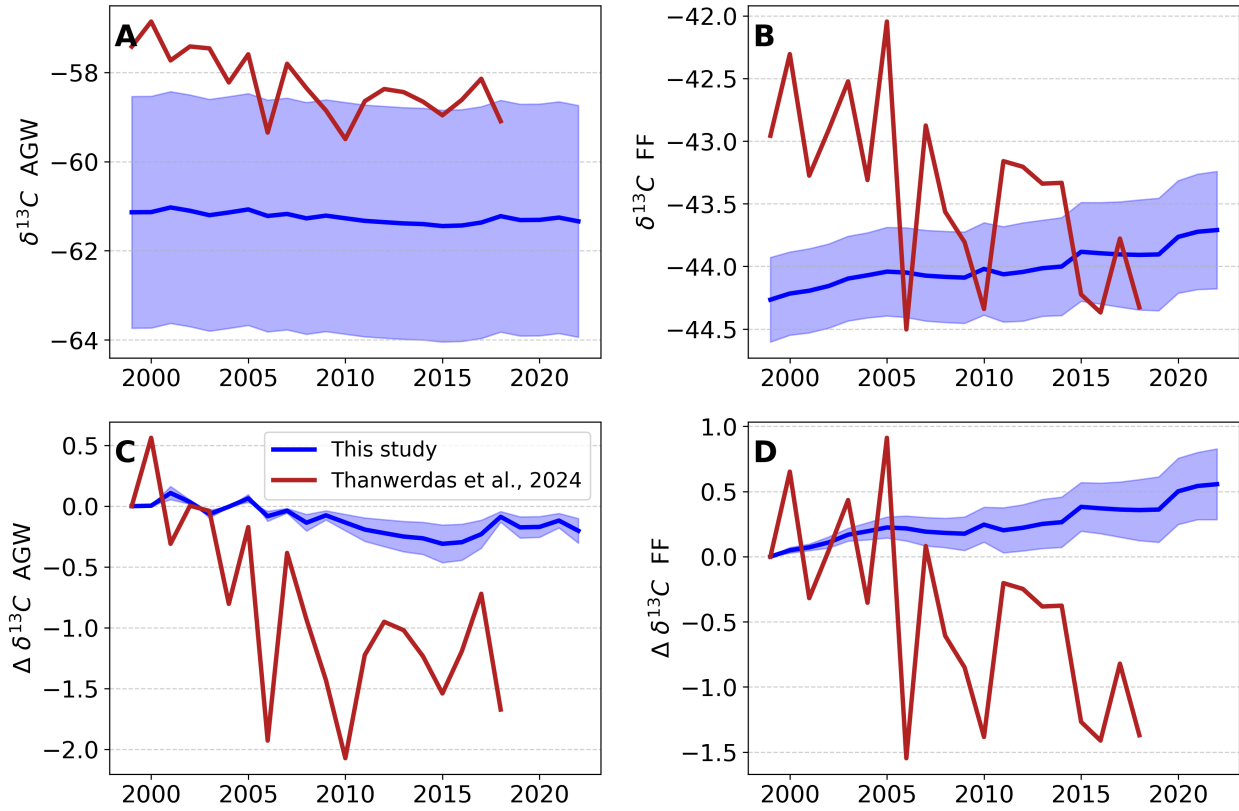


Fig. S12. Source signature trend comparison to (14). Global mean (A & C) agriculture and waste (AGW) and (B & D) fossil fuel (FF) $\delta^{13}\text{C}-\text{CH}_4$ from (14) (red) and this study with two sigma uncertainty windows used in our mass balance approach (blue line and shading). (A & B) Absolute source signatures and (C & D) trends relative to 1999. Global mean AGW $\delta^{13}\text{C}-\text{CH}_4$ in our study was calculated using the weighting approach for global mean microbial $\delta^{13}\text{C}-\text{CH}_4$, limited to the landfill, rice, and ruminant subcategories (Methods section 2.2).

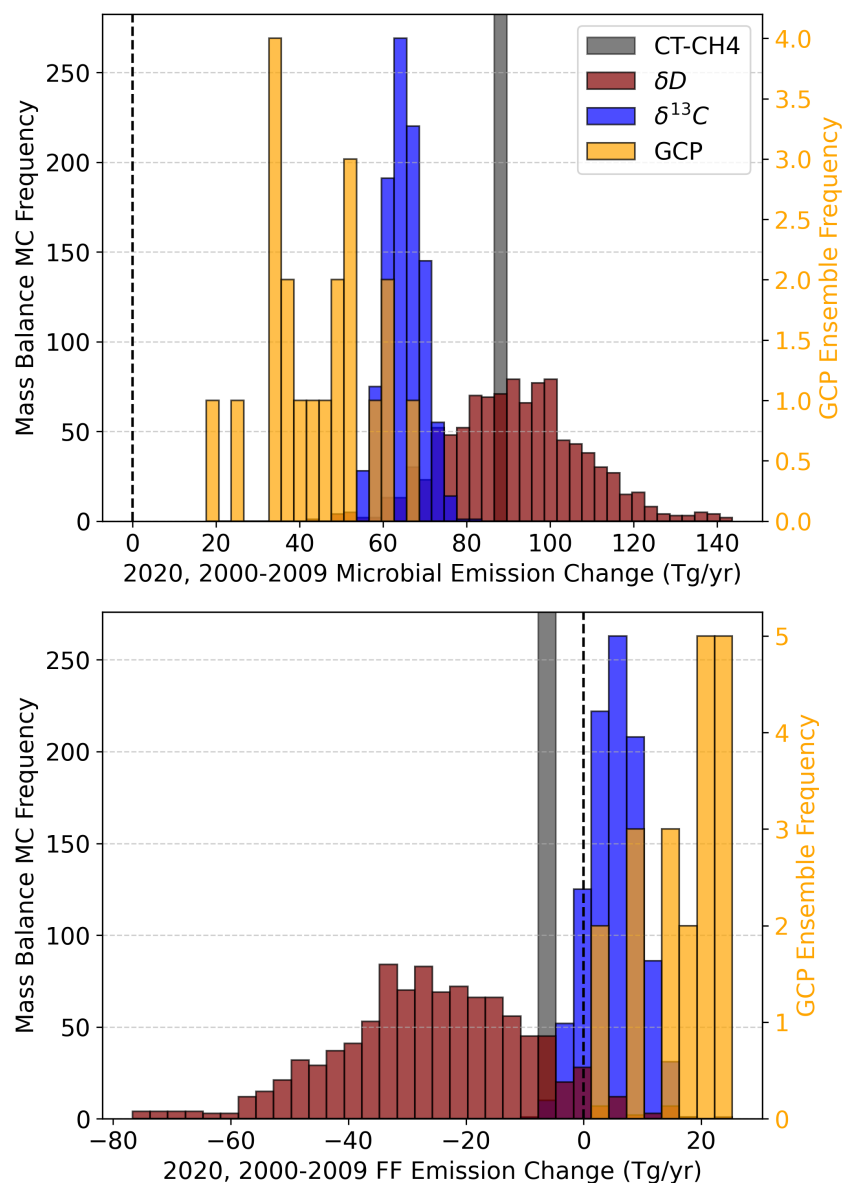


Fig. S13. Comparison to other studies. Histogram showing the unsmoothed change in microbial (A) and fossil fuel (B) emissions from the 2000-2009 average (2005 to 2009 for the δD -CH₄ mass balance) to 2020 for the Monte Carlo mass balance for $\delta^{13}C$ -CH₄ (blue bars) and δD -CH₄ (maroon bars), the CT-CH₄ posterior (gray bar) (2), and the Global Carbon Project (GCP) ensemble members (yellow bars) (15). The vertical black dashed lines indicate no change.

Supporting Tables:

Lab	# of MBL Sites	Time frame	N	Umezawa MPI Offset (‰)	Dasgupta MPI Offset (‰)
INSTAAR	10	2005 – 2009	1118	- 11.5 ± 1.5 *	- 10.9 ± 2.2 *
MPI BGC	6	2010 – 2024	2252	-	-
UU IMAU	5	1988 – 2024	1053	+ 4.2 ± 1.2	+ 2.4 ± 1.6
TU NIPR	2	1995 – 2023	962	- 8.9 ± 1.3	- 10.8 ± 1.6

Table S1. Summary of all MBL data used to calculate global mean. From left to right, laboratory name, # of MBL measurement sites available, years that data is available, number of measurements available, and the mean and standard deviation of lab offsets from (3) with the new comparison (*), and the recent lab offset estimates from (13) between MPI and INSTAAR.

Category	Subcategory	$\delta^{13}\text{C-CH}_4$		$\delta\text{D-CH}_4$	
		N	New N	N	New N
Microbial	Wetlands	2135	1287	949	750
	Landfills/Waste	395	338	147	119
	Rice paddies	375	122	83	19
	Ruminants/Wild Animals	313	142	112	18
	Termites	36	7	1	0
	Total	3254	1896	1292	906
Fossil	Coal	1989	587	745	234
	Oil & Natural Gas	7487	1407	2498	527
	Shale Gas	1735	1089	512	115
	Total	11211	3083	3755	876
Pyrogenic	Total	964	57	42	38

Table S2. Summary of all source signature measurements for each subcategory in the updated source signature database. “New N” refers to the number of new measurements compiled by this study. New N was calculated by subtracting the number of measurements in (1) from the new total (N).

Sink	Proportion of Total Sink	¹³ C KIE	^D KIE
CH₄ + OH	0.835	1.0054 ¹	1.294 ²
CH₄ + Cl	0.035	1.066 ³	1.520 ³
Stratospheric Loss	0.06	1.003 ⁴	1.179 ⁴
Soil Oxidation	0.07	1.020 ⁵	1.083 ⁵
Net Sink	1.00	1.0082	1.281

Table S3. Summary of KIEs (Kinetic Isotope Effects, with the conventions ¹²C/¹³C and H/D respectively) for ¹³C and deuterium for specific atmospheric CH₄ sinks and the net sink. Superscripts denote the reference for each KIE: ¹ (16); ² (17); ³ (18); ⁴ (19); ⁵ (20). The total sink proportion for the chlorine (21), stratosphere (22), soil (23), and OH (24) sinks are shown.

Microbial Subcategory	$\delta^{13}\text{C-CH}_4$		$\delta\text{D-CH}_4$	
	Mean	1 σ Unc.	Mean	1 σ Unc.
Wetlands	- 62.1‰	$\pm$ 0.7‰	- 313.2‰	$\pm$ 7.4‰
Ruminants	- 65.0‰	$\pm$ 1.5‰	- 309.4‰	$\pm$ 7.2‰
Waste	- 55.1‰	$\pm$ 4.4‰	- 279.4‰	$\pm$ 27.4‰
Rice Paddies	- 60.2‰	$\pm$ 4.5‰	- 311.8‰	$\pm$ 7.3‰
Wild Animals	-65.5‰	$\pm$ 3.1‰	- 301.4‰	$\pm$ 6.8‰
Termites	- 64.8‰	$\pm$ 7.6‰	- 323.1‰	$\pm$ 8.0‰
Net Microbial	- 62.1‰	$\pm$ 1.3‰	- 304.8‰	$\pm$ 8.0‰

Table S4. Summary of microbial emission subcategory source signatures ($\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ used to calculate the global mean. Values are shown for the year 2022 as in Fig. S8. See methods section 2.2 for details on the mean and uncertainty calculations.

Category	$\delta^{13}\text{C-CH}_4$ Source signature calculations
Wetlands	Annual global mean from (4).
Landfills/Waste	Global mean of all measurements with imposed Suess Effect trend.
Rice paddies	Global mean of all measurements with imposed Suess Effect trend.
Ruminants	Global mean of all measurements with imposed Suess Effect trend.
Wild Animals	Annual global mean from (7) with extrapolated trend.
Termites	Global mean of all measurements with imposed Suess Effect trend.
Total Microbial	Subcategory global mean $\delta^{13}\text{C-CH}_4$ weighted with CT-CH ₄ posterior microbial subcategory proportions.
Coal	Country-specific $\delta^{13}\text{C-CH}_4$ map flux-weighted with EDGARv8 emissions.
ONG	Country-specific $\delta^{13}\text{C-CH}_4$ map (US, China & Canada time-varying) flux-weighted with EDGARv8 emissions.
Geologic	Assumed the same as total fossil.
Total Fossil Fuel	Coal and ONG $\delta^{13}\text{C-CH}_4$ maps flux-weighted with EDGARv8 emissions.
Biomass Burning	Time-varying $\delta^{13}\text{C-CH}_4$ map from (9) flux-weighted with CT-CH ₄ posterior biomass burning emissions.

Table S5. Summary of methods used to calculate global mean $\delta^{13}\text{C-CH}_4$ source signatures. The 2023 version of CT-CH₄ posterior emissions was used for flux weighting when specified. See methods section 2 for details.

Category	$\delta\text{D-CH}_4$ Source signature calculations
Wetlands	Microbial $\delta\text{D-CH}_4$ map based on regression from (5), flux-weighted with CT- CH_4 posterior wetland emissions.
Landfills/Waste	Landfill $\delta\text{D-CH}_4$ map based on new landfill-specific regression (Fig. S6), flux-weighted with CT- CH_4 posterior landfill emissions.
Rice paddies	Same methodology as for wetland $\delta\text{D-CH}_4$.
Ruminants	Same methodology as for wetland $\delta\text{D-CH}_4$.
Wild Animals	Same methodology as for wetland $\delta\text{D-CH}_4$.
Termites	Same methodology as for wetland $\delta\text{D-CH}_4$.
Total Microbial	Subcategory global mean $\delta\text{D-CH}_4$ weighted with CT- CH_4 posterior subcategory proportions.
Coal	Country-specific $\delta\text{D-CH}_4$ map flux-weighted with EDGARv8 emissions.
ONG	Country-specific $\delta\text{D-CH}_4$ map (US and China time-varying) flux-weighted with EDGARv8 emissions.
Geologic	Assumed the same as total fossil.
Total Fossil Fuel	Coal and ONG $\delta\text{D-CH}_4$ maps flux-weighted with EDGARv8 emissions.
Biomass Burning	Biomass Burning $\delta\text{D-CH}_4$ map based on regression from (25), flux-weighted with CT- CH_4 posterior biomass burning emissions.

Table S6. Same as table S5 but for $\delta\text{D-CH}_4$ source signatures.

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